

Tree-Based Machine Learning Regression Models for Predicting Marathon Race Times

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Abstract—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Phasellus vel nulla gravida, finibus diam non, vestibulum urna. Nulla facilisi. Vivamus tempor ac metus eu facilisis. Cras imperdiet efficitur luctus. Pellentesque dignissim, odio non laoreet sodales, ligula nunc sagittis risus, vel volutpat sapien velit in neque. Proin maximus iaculis est, nec tristique risus venenatis a. Maecenas maximus neque vel odio volutpat mattis a vel lorem. Nullam ac dui bibendum, maximus massa vel, rutrum sapien.

Index Terms—Lorem, Ipsum, Dolor, Sit, Amet

I. INTRODUCTION

In recent years, various health benefits have been associated with regular physical activity [1]. As a result, long-distance endurance sports such as marathons, ultra-marathons, and triathlons have increased in popularity [2]–[4]. Esteve-Lanao *et al.* [2] state that participation in marathon events by recreational runners has become immensely popular worldwide. Additionally, wearable tracking devices have become a common necessity [5] [6] as they provide real-time sport-related data (e.g., pace, heart rate). This, in turn, helps individuals achieve a target performance, either during training or a race.

Predicting a marathon race time (MRT) is highly important and provides various entities with insightful information, but some researchers [2] [5] [7] considered it a difficult task. Amongst others, the underlying reasons point toward the complexity of the sport itself, including the event landscape, unpredictable environmental conditions, different runner biometrics [7], and different tracking devices [5]. This led to the introduction of machine learning (ML) to improve accuracy, including tree-based algorithms (supervised), clustering (unsupervised), and time-series algorithms.

The purpose of this experimental study [8] is to investigate the use of tree-based ML models to predict an athlete's MRT and subsequently evaluate the comparative predictive performance of Random Forest (RF) and Extreme Gradient Boosting (XGBoost) regressors across diverse historical data. The dependent variable of this study is the predictive performance of the ML algorithms in estimating the continuous MRT value, whilst the independent variables will be the dataset and the model type (RF against XGBoost). Specifically, 12-week and 16-week training blocks of historical data will be aggregated and used for each experiment. Moreover, the control variables will be the dataset features, including activity

metrics and demographic information, and the train-test data split strategy, so that each ML regressor can make decisions on the same data. This research does not take into consideration the landscape or weather conditions, nor any anthropometric or physiological factors that might alter the result.

To achieve this, the study aims to explore the effectiveness of tree-based ML models in predicting an athlete's MRT and to identify the optimal model based on predictive performance. Furthermore, this study seeks to determine whether the number of weeks of historical data yields a quantifiable increase in the models' predictive performance and robustness. Specifically, the hypothesis of this research states that an XGBoost ML regressor will achieve higher predictive performance than an RF regressor. Additionally, it is expected that XGBoost will be most accurate when trained on larger volumes of historical activity data (16-week training block), as this will supply the model with more comprehensive data about an individual's marathon buildup.

In view of the previously mentioned factors and as illustrated in Fig. 1, a positivist philosophy will be adopted as this study is highly focused on gathering quantifiable outcomes (predictive performance) and continuous values (the MRT). A deductive approach will be implemented because a hypothesis has been established, and the study is designed to prove or disprove it.

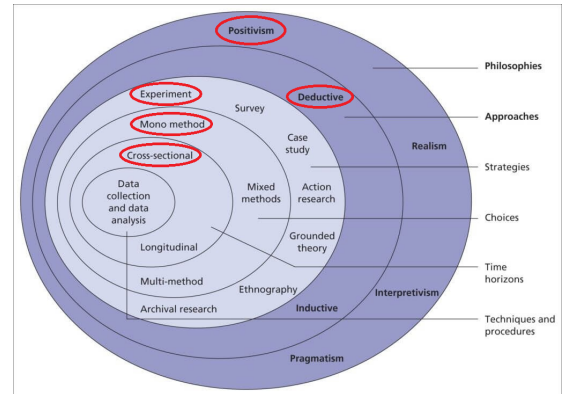


Fig. 1. The Research Onion - Adopted by [9]

In addition, an experimental strategy will be conducted,

given that the research has a dependent variable (accuracy), two independent variables (model type and timeframe of historical data), and various control variables to ensure an unbiased investigation. Thus, a mono-method quantitative research choice will be adhered to, as no surveys, interviews, or questionnaires will be held. It will depend exclusively on an open-source dataset as input and values obtained from different experiments as output. Consequently, although the dataset encompasses a period of time, a cross-sectional time horizon will be utilized. This is because the experiments will leverage a snapshot of aggregated data, rather than monitoring athletes during a specific training block.

II. LITERATURE REVIEW

The running community has grown immensely over the past few years, driving the demand for wearable tracking devices that capture real-time, sport-related data. Due to the complexity of forecasting marathon race times, predictive machine learning (ML) algorithms have become a prevalent and necessary resource for individuals in training and attaining target performances. As this technological integration has transformed how athletes strategize for major marathon races, various researchers have acknowledged and sought to improve these predictive models.

This section evaluates research methodologies adopted from a chosen set of studies and analyses the different approaches and frameworks employed by the researchers. Fig. 2 illustrates the selected set of academic studies, presenting a more comprehensive understanding of the contributions and interconnections to the predictive side of this domain. To deliver a systematic and comparative analysis, this section is based on the research onion model proposed in [9] and the research design framework proposed in [8]. Common methodological patterns will be highlighted, along with their advantages and disadvantages.

A. Instance-Based, Regression, and Deep Learning Models for Performance Prediction

With very similar objectives in mind, all six studies [10] [11] [12] [13] [14] adopted a positivist standpoint, considering an individual's performance in a marathon event a continuous, measurable, and objective outcome that can be precisely calculated by machine learning (ML) techniques. Hence, they were all distinctly deductive in nature as they validated existing hypotheses using experimental designs and historical data. Almost all studies utilized longitudinal data gathered over multiple years, except for [11]. Additionally, they entirely depend on quantitative, numerical data, distinguishing them as mono-method quantitative methodological studies. No qualitative information was gathered about an athlete's emotional state or training experience.

Both Feely *et al.* [10] and Lerebourg *et al.* [11] employed the same instance-based learning approach as a regressor to predict athlete marathon race times (MRT). The difference was that the former study coupled the k-Nearest Neighbour (KNN) model with a case-based reasoning (CBR) system,

which helped bypass the necessity of complex physiological domain models and mitigated the likelihood of unrepresentative training recommendations. The latter study compared a KNN model against an Artificial Neural Network (ANN), on the same objective. A clear difference is observed in the time horizon when choosing longitudinal [10] and cross-sectional [11] historical datasets.

Feely *et al.* [10] gathered thousands of activity records from 19,930 individuals between 2014 and 2017, and notably, due to forecasting difficulties, the runners were eligible based on their marathon performance. It included finish times between three and five hours for females and less than five hours for males, and although this resulted in better accuracy (even through visual illustrations), including sub-three-hour female athletes should have been considered, as it provides variance. On the other hand, Lerebourg *et al.* [11] investigated solely using four static inputs from 820 runners, including at least one 10-kilometer race result. These were appropriately normalized before being fed to the ANN model, aligning with the sigmoid transfer function, and to eliminate scale variations between the input variables for the KNN model, for accurate Euclidean distance calculations.

Regarding this study [10], the large-scale dataset was essential and proved highly beneficial when performing 3 separate experiments on feature variations. Although the Root Mean Square Error (RMSE) showed the models' improved performance over time, one must consider that it was not put through real-world trials. Furthermore, the dataset by Lerebourg *et al.* [11] was effective in addressing overfitting and showed that KNN outperformed the ANN model; however, incorporating a half-marathon race result from the same year could improve the model's predictive performance. This study [10] also stated that the k value was set to 15 after showing a point of error stabilization, whilst Lerebourg *et al.* [11], although having an acceptable Mean Absolute Error (MAE), set the k value to 3 without any explicit reasons. This could mean that other values could have led to better results. A common shortcoming of both studies is gender imbalance, with a 24% [10] and 17% [11] female sample population, which could result in inaccurate training recommendations and MRT prediction for this gender.

In another study by Ruiz-Mayo *et al.* [12], two regressors were evaluated and compared with a baseline ZeroR model: Linear Regression (LR) and Bagging, to assess the validity of more complex regressors for forecasting the MRT. Starting with the time horizon, the 6-week longitudinal dataset selection contradicts the researchers. It was explicitly mentioned that several months of preparation are required to compete. Without any explanation for this period selection, it might depict uninformed decisions, as this hides the majority of the training period, which could have resulted in better model performance. Adequate filtering was held, and relevant features were engineered based on weekly, bi-weekly, and the full training window. These supply the models with significant information, which contributed towards low MAE values, showing the importance of rigorous pre-processing.

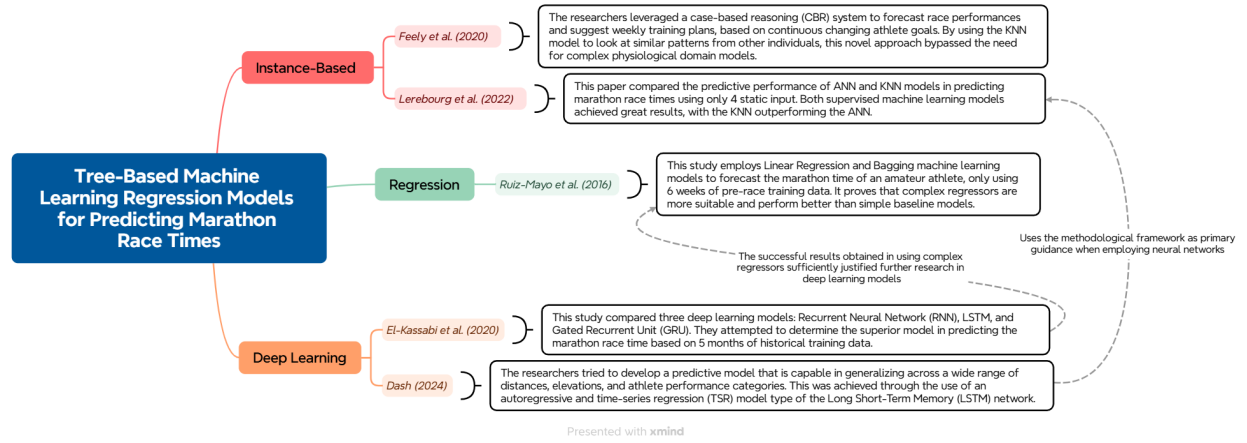


Fig. 2. Literature Map

The issue with the employed dataset was the population size, which contained only 1,000 suitable athletes from the original 3,386, coming from two major marathon events in the first half of the year, whilst there are a total of six. Therefore, generalizability could be difficult when applied to others. Moreover, using the Weka and Knime software and the standard 10-fold cross-validation, the researchers [12] developed a Bagging model of 100 combined regression trees that attained the lowest MAE of all the models. Thus, testing the algorithm with slightly more regression trees at a time could reveal better results. Nonetheless, the study still meticulously held the three separate experiments, starting with the original, filtered, and then the injury-free dataset.

Furthermore, both Dash *et al.* [13] and El-Kassabi *et al.* [14] leverage DL models for their investigations, as the former employed a regression and time-series Long Short-Term Memory (LSTM) model, while the latter compared the predictive performance among a Recurrent Neural Network (RNN), LSTM, and Gated Recurrent Unit (GRU). Although DL models were used, the objective differed. El-Kassabi *et al.* [14] focused on predicting the MRT, and Dash *et al.* [13] developed a model that forecasted an athlete's performance across a wide range of distances, elevations, and performance categories. Both studies based their experiments on longitudinal and sequential historical training logs, with an average of 6.6 years [13] and 5 months [14]. The sequential aspect is of great importance and was rightly considered due to the nature of DL models. Also, the time horizon aspect provided the algorithms with valuable information, which was reflected in the results.

Regarding the participants, Dash *et al.* [13] had one issue, as only 15 recreational runners were included in the study. Additionally, one model was developed and modified for each individual. Thus, future work would suggest gathering and testing the superior model on a larger sample population and developing a generalized LSTM predictive model, as this is not contextually feasible. This would result in more training patterns, enhancing model training, increasing scalability, and

highlighting certain flaws and possible improvements. However, it must be noted that the specialised filtering process exhibited lower Mean Absolute Percentage Error (MAPE) values, showing its significance again, as in [12]. Even the hyperparameter optimization framework (Optuna) was selected strategically, likely due to its efficiency. Overfitting was also addressed by having two regularization methods, dropout and early stopping.

Analysing the dataset in this study [14], the preparation period was more than accounted for, as mentioned in [12], and the experiments were systematically designed and executed, starting with 25 and then 50 hidden states for each model. To further enhance the dataset, the statistical three-sigma rule was adopted to eliminate outliers, yielding a highly concentrated dataset. This ensured an unbiased comparison and clear evaluation of the superior GRU model, primarily through the MAPE metric, as in [13]. A possible issue would be generalising the model, similar to the issue in [12], since it was only applied to the Boston marathon. Thus, this should be tested on other major marathons to validate its applicability. The experimental findings also showed that more hidden states contributed to slightly more precise predictions; therefore, future research should assess how MRT predictions change with gradual increments in hidden states.

III. RESEARCH METHODOLOGY

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IV. FINDINGS & DISCUSSION OF RESULTS

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V. CONCLUSION

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APPENDIX A SUPPORTING MATERIAL

You can add screen shots and statistics. Stick to essential information.

ACKNOWLEDGEMENT

You can dedicate this section for special assistance that you were given by a 3rd party. Not your mentor, relatives or questionnaire participants.

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